

0 3

Table 1 shows the names of three biological molecules and different combinations of elements.

Table 1

	Cellulose	Triglyceride	Antibody	
A	C, H, O	C, H, O	C, H, O, N	
B	C, H, O	C, H, O	C, H, O, N, S	
C	C, H, O	C, H, O, N	C, H, O, N, S	
D	C, H, O, N	C, H, O	C, H, O, N, P	

0 3 . 1

Which row identifies the correct elements in each molecule?

Tick (✓) **one** box.

[1 mark]

0 3 . 2

Describe the biochemical test used to confirm the presence of lipid in a sample.

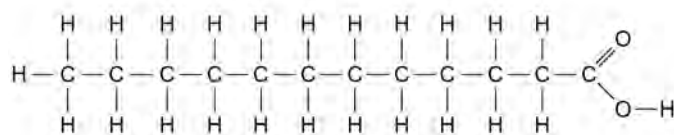
[2 marks]



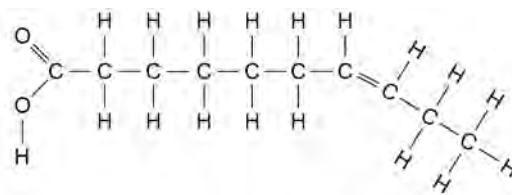
0 3 . 3 Figure 5 shows two types of fatty acids.

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Figure 5



Fatty acid A



Fatty acid B

State which fatty acid in **Figure 5** is unsaturated **and** explain your answer.

[1 mark]

Unsaturated fatty acid _____

Explanation _____

Question 3 continues on the next page

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0 3 . 4

Cholesterol is a type of lipid transported in the blood. High blood cholesterol concentrations can lead to cardiovascular disease.

Scientists studied the effectiveness of the drug cholestyramine (CA) in lowering blood cholesterol concentration.

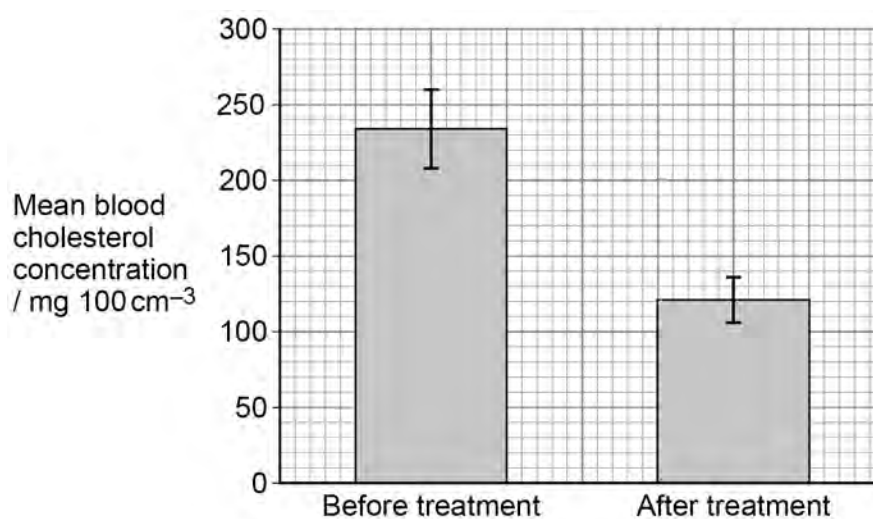
They:

- gave 10 patients CA for 1 month
- measured the blood cholesterol concentrations at the start of the study and after 1 month of treatment with CA.

Figure 6 shows the scientists' results.

The error bars represent ± 2 standard deviations from the mean, which includes over 95% of the data.

Figure 6



Use the information provided to evaluate the effectiveness of the drug CA in reducing blood cholesterol concentration.

[5 marks]

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Turn over for the next question

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